

Interactive Text-Based Games

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ABSTRACT

A one paragraph summary of your whole project. This should be complete enough for the reader to understand your goals, approach, and what you did and learned.

Keywords

Keywords are your own designated keywords separated by semicolons (";").

1. INTRODUCTION

Traditional text-based games often suffer from static narrative constraints. In these systems, player action is limited to a pre-defined "branching tree" structure. Once a player has explored the available scripted paths, the story reaches a definitive end, offering no further engagement.

This project aims to assist immersive gamers and narrative enthusiasts who seek a deeper connection with the game world and characters. For instance, some players might want to inquire about a character's hidden motivations as the story unfolds. Some players might want to express empathy or emotional responses toward certain characters.

Currently, traditional text-based game systems lack the architectural flexibility to facilitate this level of interpersonal nuance, creating a barrier between the player's desire for expression and the game's ability to respond.

2. DISCUSSION OF ISSUES

2.1 Media Issues

Platform Integration: The game will be deployed as a web-based or desktop application. The central issue is maintaining a seamless transition between the traditional text-based game and the LLM-powered chat system.

Interface Transition: The main issue will be the control scheme shift between the traditional point-and-click selection of the main game and the free-form text input of the post-game chat system.

2.2 Cognitive Issues

Attention and Distraction: The team has to come up with the interface design such that it minimizes external distractions and prevents "information overload" during text-heavy narrative segments.

Universal Accessibility: The team has to come up with the system support users with disabilities. Specifically, how will it handle requirements for adjustable text-scrolling speeds, brightness levels, and clear, high-legibility formatting?

2.3 Social Issues

Thematic Suitability: The chat system must be clearly defined and monitored. Because the LLM can generate unscripted responses, there is a risk of the AI producing content that deviates from the game's intended age rating (e.g., PG-13 vs. Mature).

Adversarial Inputs (Bad Actors): A significant social challenge is the "jailbreaking" of the LLM. Bad actors may attempt to prompt the system to generate toxic, biased, or inappropriate content. The LLM chat system has to be able to handle this issue.

3. APPROACH

Our approach involves the development of a modular narrative framework split into two distinct sub-projects. By decoupling the scripted gameplay from the dynamic dialogue system, we can provide a structured narrative foundation while offering unprecedented post-game agency.

The software will be divided into two independent applications:

1. The Scripted Text-Based Games: This application handles the core story through a traditional text-based interface. Its primary goal is to establish the world, the stakes, and the character archetypes through a series of deterministic choices. This ensures that the foundational story remains coherent and high-quality.
2. The LLM Character Dialogue System: This standalone chat application system serves as an "immersive extension." Once players have completed the scripted content, they transition to this system to engage in open-ended, natural language conversations. This addresses the player's desire for deeper inquiry and emotional expression without risking the logic of the main game.

4. PRIOR AND RELATED WORK

What has been done to help with these activities/issues before.

What techniques have been used to help in other activities that might be of value in this situation.

Remember to include references (e.g. in Section 10 below)!

5. DESIGN

What is the conceptual design of your system? This includes how will users interact with it, what it looks like, how information about the user and their tasks is captured and represented. Include figures!

6. IMPLEMENTATION

6.1 Architecture/Data Flow

This subsection might be part of the design.

It should describe the overall system design (its components) and how the components communicate (which communicate with which other components) or describe how the data needed for the system is located, preprocessed, and ingested.

6.2 Interface

Show the user interface.

Give explanations/examples of what the user will do to get their activity done (e.g. can be a scenario or a description of alternative paths the user will follow.

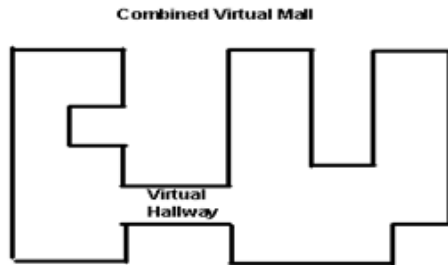


Figure 1. Insert caption to place caption below figure.

Figure 1: Enter Caption

6.3 Interactive Real-time Conversation

The conversational module is powered by the Qwen3 [1], optimized for instruction following and dialogue coherence. To address the computational bottlenecks of autoregressive generation, we implement the backend serving layer using the vLLM engine. vLLM is utilized specifically for its efficient memory management algorithm, *PagedAttention*, which treats the Key-Value (KV) cache as non-contiguous memory blocks. This approach maximizes the GPU memory utilization U_{mem} and enhances the inference throughput $R_{\text{tokens/s}}$. Consequently, the system achieves sub-second latency for inference requests, preserving the real-time nature of the user-agent interaction while maintaining high generation quality.

7. EVALUATION

Describe how you would evaluate your prototype.
Who are the participants?
What do they do?
What data will you collect?
How will you analyze the data?

8. CONCLUSIONS AND FUTURE WORK

What did you accomplish in solving the original problem?

What did you learn in the process?

What would you do to extend this if you had more time?

9. WORK PLAN

The team will be divided into two sub teams. One team will be responsible for developing LLM system. The other will work on the text-based game. The tasks of each team members are as follows:

1. Text-based game team
 - (a) Shuning Gu: Design the text game storyline and help complete the basic framework code.
 - (b) Praewa Pitiphat: Making presentation slides and provide static game assets (characters, background).
 - (c) Siyuan Yang: Responsible for the project report (writing, proofreading, editing).
2. LLM team
 - (a) Mingyang Wu: Implement LLM Engine for real-time conversation.
 - (b) Xiangbo Gao: Deploy LLM & Design LLM Agent Interaction work flow.
 - (c) Sueray Wang: Animate the assets into live2D models with physical simulation techniques for the LLM chat system.

References

- [1] A. Yang, A. Li, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Gao, C. Huang, C. Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

1. APPENDIX A: USE OF AI

Describe how your team used AI in your project, if at all.

2. APPENDIX B: PROMPT TEMPLATES

Provide any templates and static facts for prompts used in your project. Explain role and use if not obvious.

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